Fiscal Year:	FY 2021	**Task Last Updated:**	FY 11/08/2023
PI Name:	Johnson, Christina Ph.D.		
Project Title:	Augmenting Nutritionally-Dense Microgreens in Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	christina.m.johnson@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	NASA Postdoctoral Fellow		
PI Address 2:	KSC-UBA		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32899	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NASA Postdoctoral Program
Start Date:	08/26/2019	**End Date:**	02/28/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
Contact Email:	gioia.massa@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 02/28/23 per PI (Ed., 1/9/23). PI also notes that administration of NPP was moved from USRA to ORAU. NOTE: End date changed to 8/26/2022 per G. Massa/KSC; original end date was 8/26/2021. (Ed., 11/30/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wheeler, Raymond Ph.D. (MENTOR/ NASA Kennedy Space Center)		
Grant/Contract No.:	NASA Postdoctoral Program		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA Postdoctoral Program Fellow For long-duration spaceflight missions, providing adequate nutritional needs of the crew is a documented risk. This can be mitigated with dietary supplementation of nutrient-dense young vegetables called microgreens. By optimizing growing conditions such as lighting and fertilizer application within a simulated microgravity environment we can augment the nutrient density of microgreens, producing crops that are specifically designed to meet crew nutritional needs in the microgravity environment of spaceflight. In this proposal we will further screen plants in the Brassicaceae family that have been selected by the USDA (United States Department of Agriculture) for nutritional content, and select for plants that show optimal growth in spaceflight-analog conditions, including the use of multi-axis clinostats to simulate a microgravity setting. We will adapt an existing method of fertilizer application to one applicable to microgravity. We propose to quantify cell wall structural changes that accompany enhanced growth through high-throughput glycomics and compare the results with previous spaceflight studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Ed. Note: Compiled from Final Report submitted to NASA (Ed., 11/8/23) Plasma for Agriculture I participated in a multidisciplinary collaboration between plasma physicists and biologists at NASA Kennedy Space Center (KSC). To kick off our work together, we wrote a literature review: Reviewing Plasma Seed Treatments for Advancing Agriculture Applications on Earth and Into the Final Frontier. I conducted preliminary studies growing microgreens in plasma treated water, and germination studies on seeds that had been exposed to various plasma treatments for seed sanitization. I analyzed data from these experiments, including comparing results of plasma treatments with microbial loads and germination rates to determine trends that we could pursue with additional studies. We wrote this work into a scientific publication: Plasma sanitization of cherry belle radish seeds for Space agricultural applications. [Ed. Note: See Bibliography.] Simulated Differential Gravity Studies I conducted many studies with microgreens in elevated carbon dioxide and various environments – including simulated microgravity (~0 g), Lunar (1/6 g), and Martian (1/3 g) gravity. We grew microgreens in trays in International Space Station (ISS)-like conditions (i.e., similar humidity, temperature, and CO2 as on the ISS). Each tray was illuminated by a NASA Vegetable Production System (Veggie) analog with LED light cap. These tray studies served as a testbed, allowing me to look at many different species and varieties of microgreens to compare their growth and development. The work in simulated differential gravity was conducted in coordination with the KSC Microgravity Simulation Support Facility (MSSF), using clinostats and Random Positioning Machines (RPMs). All clinostats work on the principle of negating the directional influence of gravity by continually rotating the plants. [Ed. Note: See Bibliography.] We designed and verified a specialized hardware to grow microgreens as a crop on the RPM. Most of these tests used radish (Raphanus sativus cv. Cherry Belle) seedlings. During many of our trials there were a range of challenges to overcome. Many of our attempted experiments were stopped short by various equipment malfunctions. These included power outages that caused the rotations to stop mid-study, fluids leaking, lighting differences, and eventually hardware wearing out and shifting with the strain of continuous use. Even with challenges along the way, we gathered the data from many successful runs and were able to compare our growth across the simulated gravities. We also collected samples for future transcriptomics studies. This work is ongoing and will be published when possible. Harvesting Hardware Development and Testing We worked with engineers to develop hardware that would simultaneously grow and harvest microgreens in a microgravity environment, then tested this hardware in a parabolic flight. As a team, we refined the hardware types, then successfully grew plants in our hardware and took our hardware on a parabolic flight to evaluate the hardware and harvesting methods in various gravities. We took surveys in-flight and post-flight; we gathered biomass measurements of our harvests. Manuscript in progress: Design, build and testing of hardware to safely harvest microgreens in microgravity. [Ed. Note: See Bibliography.]
Bibliography Type:	Description: (Last Updated: 11/08/2023)
Articles in Peer-reviewed Journals	Barker R, Kruse CPS, Johnson C, Saravia-Butler A, Fogle H, Chang HS, Trane RM, Kinscherf N, Villacampa A, Manzano A, Herranz R, Davin LB, Lewis NG, Perera I, Wolverton C, Gupta P, Jaiswal P, Reinsch SS, Wyatt S, Gilroy S. "Meta-analysis of the spaceflight and microgravity response of the Arabidopsis plant transcriptome." npj Microgravity. 2023 Mar 20;9(1):21. https://doi.org/10.1038/s41526-023-00247-6 ; PMID: 36941263; PMCID: PMC10027818 , Mar-2023
Articles in Peer-reviewed Journals	Teng Z, Luo Y, Pearlstein DJ, Zhou B, Johnson CM, Mowery J, Wang Q, Fonseca JM. "Agarose hydrogel composite supports microgreen cultivation with enhanced porosity and continuous water supply under terrestrial and microgravitational conditions." Int J Biol Macromol. 2022 Nov 1;220:135-46. https://doi.org/10.1016/j.ijbiomac.2022.08.046 , Nov-2022
Articles in Peer-reviewed Journals	Teng Z, Luo Y, Pearlstein DJ, Wheeler RM, Johnson CM, Wang Q, Fonseca J. "Microgreens for home, commercial, and space farming: A comprehensive update of the most recent developments." Annu Rev Food Sci Technol. 2023 Mar 27:14:539-562. Review. https://doi.org/10.1146/annurev-food-060721-024636 ; PMID: 36525689 , Mar-2023
Articles in Peer-reviewed Journals	Gott RP, Engeling KW, Johnson CM, Boles HO, Brown VC, Hummerick M, Massa GD, Meier AJ. "Plasma sanitization of cherry belle radish seeds for Space agricultural applications." Plasma Res Express. 2022 Apr 14;4(2):025001. http://dx.doi.org/10.1088/2516-1067/ac64ee , Apr-2022

Articles in Peer-reviewed Journals	Poulet L, Engeling K, Hatch T, Stahl-Rommel S, Velez Justiniano Y-A, Castro-Wallace S, Bunchek J, Monje O, Hummerick M, Khodadad CLM, Spencer LE, Pechous J, Johnson CM, Fritsche R, Massa GD, Romeyn MW, O'Rourke AE, Wheeler RW. "Large-scale crop production for the Moon and Mars: Current gaps and future perspectives." Front Astron Space Sci. 2022 Feb 4;8:733944. https://doi.org/10.3389/fspas.2021.733944 , Feb-2022
Articles in Peer-reviewed Journals	Johnson CM, Boles HO, Spencer LE, Poulet L, Romeyn M, Bunchek JM, Fritsche R, Massa GD, O'Rourke A, Wheeler RM. "Supplemental food production with plants: A review of NASA research." Front Astron Space Sci. 2021 Nov 10;8:734343. Mini review. https://doi.org/10.3389/fspas.2021.734343 , Nov-2021
Articles in Peer-reviewed Journals	Meier A, Essumang D, Hummerick M, Johnson C, Kruger M, Massa G, Engeling K. "Reviewing plasma seed treatments for advancing agriculture applications on Earth and into the final frontier." Gravit Space Res. 2021 Dec 23;9(1):133-58. Review. https://doi.org/10.2478/gsr-2021-0011 , Dec-2021
Articles in Peer-reviewed Journals	Boles HO, Poulet L, Johnson CM, Torres JJ, Koss LL, Spencer LE, Massa GD. "Design, build, and testing of hardware to safely harvest microgreens in microgravity." Gravit Space Res. 2023 Oct 30;11(1):1-14. https://doi.org/10.2478/gsr-2023-0001 , Oct-2023
Articles in Peer-reviewed Journals	Zhang Y, Richards JT, Hellein JL, Johnson CM, Woodall J, Sorenson T, Neelam S, Ruby AMJ, Levine HG. "NASA's ground-based microgravity simulation facility." Methods Mol Biol. 2022;2368:281-99. (Plant Gravitropism: Methods and Protocols.). https://doi.org/10.1007/978-1-0716-1677-2_18 ; PMID: 34647262 , Jan-2022